

Luca Nunziante

Tokyo, Japan · luca.nunziante1999@gmail.com
LinkedIn · [GitHub](#) · [Google Scholar](#)

Robot learning researcher working on Vision-Language-Action (VLA) models and imitation/reinforcement learning for manipulation, with hands-on experience taking policies from simulation to real-world deployment on physical robots. Comfortable across the full stack: low-level control, teleoperation and data collection, model training, and deployment infrastructure.

Research Experience

Jun 2024 – **Robot Learning Researcher**, *Araya Inc., Tokyo, Japan*.

- Current
 - Develop imitation learning and reinforcement learning pipelines for robot manipulation built on VLA models.
 - Built end-to-end training infrastructure for IL and RL on custom datasets, improving manipulation success rate by iterating and fixing edge cases encountered during deployment.
 - Implemented low-level controllers (operational-space, CBF-based) and teleoperation interfaces enabling safe data collection and reliable real-world policy execution. (Related Publication: *Improving Low-Cost Teleoperation: Augmenting GELLO with Force*)
 - Built custom MuJoCo simulation environments and the deployment infrastructure that transfers trained policies to physical hardware

Dec 2023 – **Visiting Research Student**, *Tohoku University, Sendai, Japan*.

- Apr 2024
 - Developed compliant control methods for robotic assembly tasks and perception systems for lunar applications.
 - Built a dual-arm solar-panel assembly pipeline (MSc thesis project) that led to a peer-reviewed publication (SII 2025).

Programming Knowledge

Languages Python, C++, MATLAB, $\text{\LaTeX}$.

Libraries PyTorch, Mujoco, ROS, Eigen.

Tools git, Isaac Gym, Gazebo.

Publications

[†] Equal contribution.

2026 H. Douglas[†], M. Di Vincenzo[†], R.F.J. Dossa[†], **L. Nunziante[†]**, S. Sujit, and K. Arulkumaran, Levels of shared autonomy in brain-robot interfaces: enabling multi-robot multi-human collaboration for activities of daily living, *Frontiers in Human Neuroscience*, 19:1718713, 2026.

Project website: <https://coopopen.github.io/>

Under review S. Sujit[†], **L. Nunziante[†]**, R.F.J. Dossa, M. Di Vincenzo, and K. Arulkumaran, When One Step Is Enough: An Empirical Study of Euler Integration in Flow-Matching Policies, Under review.

- 2026 S. Zhang, R.F.J. Dossa, **L. Nunziante**, M. Di Vincenzo, and K. Arulkumaran, An Augmented Reality Brain-Robot Interface for Generalist Robot Arm Manipulation, Accepted at IEEE RO-MAN 2026.
Project website [WIP]: <https://ar-bri-manip.github.io/>
- 2025 **L. Nunziante**, K. Uno, G.H. Diaz, S. Santra, A. De Luca, K. Yoshida, Assembling solar panels by dual robot arms towards full autonomous lunar base construction, *2025 IEEE/SICE International Symposium on System Integration*, pp. 1497–1502, Munich, GER.
Project video: <https://www.youtube.com/watch?v=ZQTdP5nwqpQ>
- 2025 S. Sujit[†], **L. Nunziante**[†], D.O. Lillrank, R.F.J. Dossa, and K. Arulkumaran, Improving Low-Cost Teleoperation: Augmenting GELLO with Force, *2025 IEEE/SICE International Symposium on System Integration*, pp. 747–752, Munich, GER.
- 2024 G. Gravina[†], **L. Nunziante**[†], A. De Luca, Energy-based Residual for Collision Detection Using a Velocity Observer, *6th Italian Conference on Robotics and Intelligent Machines*, October 2024, Rome, ITA.
Project video: <https://www.youtube.com/watch?v=v73nNGPyUYQ>

Research Projects

- Nov 2025 **Resolving Uncertainty in Robotic Manipulation with Vision-Language Models and Conformal Prediction.**
A Vision Language Model is employed for robot planning, and response uncertainty is quantified using Conformal Prediction. The resulting plans are executed using an imitation learning policy. A user interface was also developed to facilitate interaction with a real robot.

Education

- 2021–2024 **MSc in Artificial Intelligence and Robotics**, *Sapienza University of Rome*, Rome, Italy, *110/110 cum laude*.
Thesis: “Autonomous Assembly of Solar Panels by Dual Robot Arms in the Remote Space” (Supervisors: Prof. A. De Luca, Prof. K. Yoshida).
- 2018–2021 **BSc in Electronic and Computer Engineering**, *University of Campania Luigi Vanvitelli*, Caserta, Italy, *110/110 cum laude*.
Thesis: “Object detection with Neural Networks for robotics applications: training methods compared” (Supervisor: Prof. C. Natale).

Languages

- Italian **Mother tongue**
English **Professional working proficiency**
Japanese **Conversational**